# MÁXIMO RETAMAL ROJAS

Computer Science Engineering, Degree Candidate | Universidad de Chile

Las Condes, Santiago, Chile | +56 9 7382 1580 | maximo.retamal.r@gmail.com | github.com/maximo1400

## PROFESSIONAL PROFILE

Computer Science Engineering degree candidate at Universidad de Chile, specializing in software engineering, Data Science, Machine Learning, and distributed architectures. Proven experience building real-time data pipelines, integrating Affective Computing and Brain-Computer Interfaces (BCI) into 3D environments (Unity/Python), full-stack web development (Django/React), and cloud deployments (AWS, Docker). Passionate about technology, cybersecurity, and solving complex engineering challenges.

## PROFESSIONAL EXPERIENCE & PROJECTS

**Bachelor's Thesis / Capstone Engineering Project** | **Universidad de Chile** *August 2025 – 2026 • Santiago, Chile*

- Designed and implemented a Python pipeline for real-time ingestion and filtering of electroencephalographic (EEG - Emotiv EPOC+) signals and processing of biometric datasets (DREAMER).
- Extracted frequency-domain features (Band Power, FFT, Butterworth IIR filters, DC removal) and trained ML models (SVM, Random Forest) to classify emotional states (Valence & Arousal).
- Developed temporal smoothing modules (circular buffers, EMA) and asynchronous streaming via TCP sockets to Unity 3D (C#), dynamically adapting the graphical environment (audio, lighting, and interactive guidance).

**BCI & Machine Learning Research Project** | **Universidad de Chile** *2023 – 2024 • Santiago, Chile*

- Developed a multi-threaded Python desktop application (Tkinter, Emotiv Cortex API) for real-time EEG telemetry acquisition (Emotiv EPOC+) and synchronized visual stimuli for mental command training.
- Designed experimental protocols and conducted empirical trials with 30 participants, capturing and structuring brainwave datasets across 7 distinct mental commands.
- Built non-linear dimensionality reduction pipelines using Autoencoders (Deep Learning) and PCA, evaluating clustering techniques to accelerate cross-user BCI transfer learning.

**Software & Data Science Engineering Intern** | **Perceptron SpA** *December 2022 – January 2023 • Santiago, Chile*

- Developed a Django/React web dashboard for production data management and predictive weight/slaughter models (Agrosuper), implementing asynchronous task queue tracking in backend and database.
- Built Python data transformation pipelines (hierarchical structures & dictionaries) in collaboration with mathematical engineers to feed linear optimization models for raw material recipes (**Cecinamix**).
- Refactored price elasticity calculation engines (**Revenew**) to compute store-branch-level metrics, and managed cloud deployments on AWS (EC2, Route 53), Docker, and Nginx.

**Software Engineering Intern** | **Wisetrack Corp** *January 2020 – March 2020 • Santiago, Chile*

- Translated SQL Server stored procedures to PostgreSQL for cleaning and structuring telemetry alerts in the **Third Eye** driver safety platform.
- Programmed a Python ETL daemon on a Linux VM to extract, transform, and replicate telemetry data into PostgreSQL and MySQL every 10 seconds.
- Stabilized and modularized facial recognition microservices (**MOBICUA**), integrated AWS Rekognition, evaluated IBM Watson AI services, and built analytics dashboards in Qlik Sense.

## EDUCATION

**Bachelor of Science & Engineering Degree in Computer Science** *2016 – 2026 (Degree Candidate) • Santiago, Chile*
Universidad de Chile — Faculty of Physical and Mathematical Sciences (FCFM)

## TECHNICAL SKILLS

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|---|---|
| **Programming Languages:** | Python, SQL, C#, JavaScript, C, Java, HTML/CSS. |
| **ML & Data Science:** | Pandas, NumPy, BCI/EEG, SVM, Random Forest, Autoencoders, PCA, Clustering. |
| **Web & 3D Development:** | Django, React, Vue.js, TCP Sockets, Unity 3D (C#), Unity 6, Tailwind CSS. |
| **Databases:** | PostgreSQL, MySQL, SQL Server, SQLite, MongoDB. |
| **Cloud & DevOps:** | AWS (EC2, Route 53, Rekognition), Docker, Nginx, Linux (Daemons, Shell), Windows, Raspberry Pi. |
| **Languages:** | Spanish (Native), English (Advanced). |